(g) The reduction potential of permanganate ion (+7 oxidation state) in acidic aqueous solution is given by

$$MnO_4^-(aq) + 8\ H_3O^+(aq) + 5\ e^- \longrightarrow Mn^{2+}(aq) + 12\ H_2O(\ell) \qquad E^o = 1.491\ V$$

whereas that of the analogous fifth-period species, pertechnetate ion, is

$$TcO_4^-(aq) + 8\ H_3O^+(aq) + 5\ e^- \longrightarrow Tc^{2+}(aq) + 12\ H_2O(\ell) \qquad E^o = 0.500\ V$$

Which is the stronger oxidizing agent, permanganate ion or pertechnetate ion?

(h) A galvanic cell is made from two half-cells. In the first, a platinum electrode is immersed in a solution at pH 2.00 that is 0.100 M in both MnO_4^- and Mn^{2+}. In the second, a zinc electrode is immersed in a 0.0100 M solution of $Zn(NO_3)_2$. Calculate the cell voltage that will be measured.

Answers

(a) Mn appears at the cathode and MnO_2 at the anode.

(b)

$$\begin{array}{rl} Mn(OH)_2(s) + OH^-(aq) \longrightarrow Mn(OH)_3(s) + e^- & \text{(oxidation)} \\ O_2(g) + 2\ H_2O(\ell) + 4\ e^- \longrightarrow 4\ OH^-(aq) & \text{(reduction)} \\ \hline 4\ Mn(OH)_2(s) + O_2(g) + 2\ H_2O(\ell) \longrightarrow 4\ Mn(OH)_3(s) & \end{array}$$

$$\Delta E^o = 0.401 - (-0.40) = 0.80\ V$$

(c) $K = 1 \times 10^{54}$

(d) Mn^{3+} will spontaneously oxidize I^-.

$$2\ Mn^{3+}(aq) + 2\ I^-(aq) \longrightarrow 2\ Mn^{2+}(aq) + I_2(s) \qquad K = 9 \times 10^{32}$$

(e) 3.8×10^3 J

(f) 0.85 g Zn is oxidized.

(g) Permanganate ion

(h) 2.12 V

PROBLEMS

Electrochemical Cells

1. Diagram the following galvanic cell, indicating the direction of flow of electrons in the external circuit and the motion of ions in the salt bridge.

$$Pt(s)|Cr^{2+}(aq), Cr^{3+}(aq)||Cu^{2+}(aq)|Cu(s)$$

Write a balanced equation for the overall reaction in this cell.

2. Diagram the following galvanic cell, indicating the direction of flow of electrons in the external circuit and the motion of ions in the salt bridge.

$$Ni(s)|Ni^{2+}(aq)||HCl(aq)|H_2(g)|Pt(s)$$

Write a balanced equation for the overall reaction in this cell.

3. A quantity of electricity equal to 6.95×10^4 C passes through an electrolytic cell that contains a solution of $Sn^{4+}(aq)$ ions. Compute the maximum chemical amount, in moles, of Sn(s) that can be deposited at the cathode.

4. A quantity of electricity equal to 9.263×10^4 C passes through a galvanic cell that has an Ni(s) anode. Compute the maximum chemical amount, in moles, of $Ni^{2+}(aq)$ that can be released into solution.

5. A galvanic cell is constructed that has a zinc anode immersed in a $Zn(NO_3)_2$ solution and a platinum cathode immersed in an NaCl solution equilibrated with $Cl_2(g)$ at 1 atm and 25°C. A salt bridge connects the two half-cells.
   (a) Write a balanced equation for the cell reaction.
   (b) A steady current of 0.800 A is observed to flow for a period of 25.0 minutes. How much charge passes through the circuit during this time? How many moles of electrons is this charge equivalent to?
   (c) Calculate the change in mass of the zinc electrode.
   (d) Calculate the volume of gaseous chlorine generated or consumed as a result of the reaction.

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6. A galvanic cell consists of a cadmium cathode immersed in a $CdSO_4$ solution and a zinc anode immersed in a $ZnSO_4$ solution. A salt bridge connects the two half-cells.
   (a) Write a balanced equation for the cell reaction.
   (b) A current of 1.45 A is observed to flow for a period of 2.60 hours. How much charge passes through the circuit during this time? How many moles of electrons is this charge equivalent to?
   (c) Calculate the change in mass of the zinc electrode.
   (d) Calculate the change in mass of the cadmium electrode.

7. An acidic solution containing copper ions is electrolyzed, producing gaseous oxygen (from water) at the anode and copper at the cathode. For every 16.0 g of oxygen generated, 63.5 g of copper plates out. What is the oxidation state of the copper in the solution?

8. Michael Faraday reported that passing electricity through one solution liberated 1 mass of hydrogen at the cathode and 8 masses of oxygen at the anode. The same quantity of electricity liberated 36 masses of chlorine at the anode and 58 masses of tin at the cathode from a second solution. What were the oxidation states of hydrogen, oxygen, chlorine, and tin in these solutions?

9. Liquid potassium chloride, $KCl(\ell)$, is decomposed in an electrolytic cell to form potassium and chlorine. Liquid KCl consists of K^+ and Cl^- ions.
   (a) Write balanced equations for the half-cell reactions at the anode and at the cathode, and for the overall cell reaction.
   (b) If a current of 2.00 A is passed through the cell for a period of 5.00 hours, calculate the mass of metal deposited and of gas liberated.

10. In the Hall–Héroult process for the electrolytic production of aluminum, Al^{3+} ions from Al_2O_3 dissolved in molten cryolite (Na_3AlF_6) are reduced to $Al(l)$ while carbon (graphite) is oxidized to CO_2 by reaction with oxide ions.
   (a) Write balanced equations for the half-reactions at the anode and at the cathode, and for the overall cell reaction.
   (b) If a current of 50,000 A is passed through the cell for a period of 24 hours, what mass of aluminum will be recovered?

Cell Potentials and the Gibbs Free Energy

11. A $Ni|Ni^{2+}||Ag^+|Ag$ galvanic cell is constructed in which the standard cell potential is 1.03 V. Calculate the free energy change at 25°C when 1.00 g of silver plates out, if all concentrations remain at their standard value of 1 M throughout the process. What is the maximum electrical work done by the cell on its surroundings during this experiment?

12. A $Zn|Zn^{2+}||Co^{2+}|Co$ galvanic cell is constructed in which the standard cell potential is 0.48 V. Calculate the free energy change at 25°C per gram of zinc lost at the anode, if all concentrations remain at their standard value of 1 M throughout the process. What is the maximum electrical work done by the cell on its surroundings during this experiment?

13. A galvanic cell is constructed in which a $Br_2|Br^+$ half-cell is connected to a $Co^{2+}|Co$ half-cell.
   (a) By referring to Appendix E, write balanced chemical equations for the half-reactions at the anode and the cathode and for the overall cell reaction.
   (b) Calculate the cell potential, assuming that all reactants and products are in their standard states.

14. A galvanic cell is constructed in which a $Pt|Fe^{2+}\ Fe^{3+}$ half-cell is connected to a $Cd^{2+}|Cd$ half-cell.
   (a) Referring to Appendix E, write balanced chemical equations for the half-reactions at the anode and the cathode and for the overall cell reaction.
   (b) Calculate the cell potential, assuming that all reactants and products are in their standard states.

15. In a galvanic cell, one half-cell consists of a zinc strip dipped into a 1.00 M solution of $Zn(NO_3)_2$. In the second half-cell, solid indium adsorbed on graphite is in contact with a 1.00 M solution of $In(NO_3)_3$. Indium is observed to plate out as the galvanic cell operates, and the initial cell potential is measured to be 0.425 V at 25°C.
   (a) Write balanced equations for the half-reactions at the anode and the cathode.
   (b) Calculate the standard reduction potential of an $In^{3+}|In$ half-cell. Consult Appendix E for the reduction potential of the $Zn^{2+}|Zn$ electrode.

16. In a galvanic cell, one half-cell consists of gaseous chlorine bubbled over a platinum electrode at a pressure of 1.00 atm into a 1.00 M solution of NaCl. The second half-cell has a strip of solid gallium immersed in a 1.00 M $Ga(NO_3)_3$ solution. The initial cell potential is measured to be 1.918 V at 25°C, and as the cell operates, the concentration of chloride ion is observed to increase.
   (a) Write balanced equations for the half-reactions at the anode and the cathode.
   (b) Calculate the standard reduction potential of a $Ga^{3+}|Ga$ half-cell. Consult Appendix E for the reduction potential of the $Cl_2|Cl^-$ electrode.

17. Would you expect powdered solid aluminum to act as an oxidizing agent or as a reducing agent?

18. Would you expect potassium perchlorate, KClO4(*aq*), in a concentrated acidic solution to act as an oxidizing agent or as a reducing agent?

19. Bromine is sometimes used in place of chlorine as a disinfectant in swimming pools. If the effectiveness of a chemical as a disinfectant depends solely on its strength as an oxidizing agent, do you expect bromine to be better or worse than chlorine as a disinfectant, at a given concentration?

20. Many bleaches, including chlorine and its oxides, oxidize dye compounds in cloth. Predict which of the following will be the strongest bleach at a given concentration and pH 0: $NaClO_3(aq)$, $NaClO(aq)$, $Cl_2(aq)$. How does the strongest chlorine-containing bleach compare in strength with ozone, $O_3(g)$?

21. Suppose you have the following reagents available at pH 0, atmospheric pressure, and 1 M concentration:

$$Co(s),\ Ag^+(aq),\ Cl^-(aq),\ Cr(s),\ BrO_3^-(aq),\ I_2(s)$$

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(a) Which is the strongest oxidizing agent?
(b) Which is the strongest reducing agent?
(c) Which reagent will reduce $Pb^{2+}(aq)$ while leaving $Cd^{2+}(aq)$ unreacted?

22. Suppose you have the following reagents available at pH 0, atmospheric pressure, and 1 M concentration:

$$Sc(s),\ Hg_2^{2+}(aq),\ Cr_2O_7^{2-}(aq),\ H_2O_2(aq),\ Sn^{2+}(aq),\ Ni(s)$$

(a) Which is the strongest oxidizing agent?
(b) Which is the strongest reducing agent?
(c) Which reagent will oxidize $Fe(s)$ while leaving $Cu(s)$ unreacted?

23. (a) Use the data from Appendix E to calculate the half-cell potential E^o for the half-reaction

$$Mn^{3+}(aq) + 3\ e^- \longrightarrow MN(s)$$

(b) Consider the disproportionation reaction

$$3\ Mn^{2+}(aq) \rightleftharpoons Mn(s) + 2\ Mn^{3+}(aq)$$

Will Mn^{2+} disproportionate in aqueous solution?

24. The following standard reduction potentials have been measured in aqueous solution at 25°C:

$$Tl^{3+} + e^- \longrightarrow Tl^{2+} \qquad E^o = -0.37\ V$$

$$Tl^{3+} + 2\ e^- \longrightarrow Tl^{+} \qquad E^o = 1.25\ V$$

(a) Calculate the half-cell potential for the half-reaction

$$Tl^{2+} + e^- \longrightarrow Tl^{+}$$

(b) Consider the disproportionation reaction

$$2\ Tl^{2+}(aq) \rightleftharpoons Tl^{3+}(aq) + Tl^{+}(aq)$$

Will Tl^{2+} disproportionate in aqueous solution?

25. The following reduction potentials are measured at pH 0:

$$BrO_3^- + 6\ H_3O^+ + 5\ e^- \longrightarrow \frac{1}{2}Br_2(\ell) + 9\ H_2O \qquad E^o = 1.52\ V$$

$$Br_2(\ell) + 2\ e^- \longrightarrow 2\ Br^- \qquad E^o = 1.065\ V$$

(a) Will bromine disproportionate spontaneously in acidic solution?
(b) Which is the stronger reducing agent at pH 0: $Br_2(\ell)$ or Br_2?

26. The following reduction potentials are measured at pH 14:

$$ClO^- + H_2O + 2\ e^- \longrightarrow Cl^- + 2\ OH^- \qquad E^o = 0.90\ V$$

$$ClO_2^- + H_2O + 2\ e^- \longrightarrow ClO^- + 2\ OH^- \qquad E^o = 0.59\ V$$

(a) Will ClO^- disproportionate spontaneously in basic solution?
(b) Which is the stronger reducing agent at pH 14: ClO^- or Cl^-?

Concentration Effects and the Nernst Equation

27. A galvanic cell is constructed that carries out the reaction

$$Pb^{2+}(aq) + 2\ Cr^{2+}(aq) \longrightarrow Pb(s) + 2\ Cr^{3+}(aq)$$

If the initial concentration of $Pb^{2+}(aq)$ is 0.15 M, that of $Cr^{2+}(aq)$ is 0.20 M, and that of $Cr^{3+}(aq)$ is 0.0030 M, calculate the initial voltage generated by the cell at 25°C.

28. A galvanic cell is constructed that carries out the reaction

$$2\ Ag(s) + Cl_2(g) \longrightarrow 2\ Ag^+(aq) + 2\ Cl^-(aq)$$

If the partial pressure of $Cl_2(g)$ is 1.00 atm, the initial concentration of $Ag^+(aq)$ is 0.25 M, and that of $Cl^-(aq)$ is 0.016 M, calculate the initial voltage generated by the cell at 25°C.

29. Calculate the reduction potential for a $Pt|Cr^{3+}, Cr^{2+}$ half-cell in which $[Cr^{3+}]$ is 0.15 M and $[Cr^{2+}]$ is 0.0019 M.

30. Calculate the reduction potential for an $I_2(s)|I^-$ half-cell in which $[I^-]$ is 1.5×10^{-6} M.

31. An $I_2(s)|I^-$ (1.00 M) half-cell is connected to an $H_3O^+|H_2$ (1 atm) half-cell in which the concentration of the hydronium ion is unknown. The measured cell potential is 0.841 V, and the $I_2|I^-$ half-cell is the cathode. What is the pH in the $H_3O^+|H_2$ half-cell?

32. A $Cu^{2+}(1.00\ M)|Cu$ half-cell is connected to a $Br_2(\ell)|Br^-$ half-cell in which the concentration of bromide ion is unknown. The measured cell potential is 0.963 V, and the $Cu^{2+}|Cu$ half-cell is the anode. What is the bromide ion concentration in the $Br_2(\ell)|Br^-$ half-cell?

33. The following reaction occurs in an electrochemical cell:

$$3\ HClO_2(aq) + 2\ Cr^{3+}(aq) + 12\ H_2O(\ell) \longrightarrow 3\ HClO(aq) + Cr_2O_7^{2-}(aq) + 8\ H_3O^+(aq)$$

(a) Calculate E^o for this cell.
(b) At pH 0, with $[Cr_2O_7^{2-}] = 0.80$ M $[HClO_2] = 0.15$ M, and $[HClO] = 0.20$ M, the cell potential is found to be 0.15 V. Calculate the concentration of $Cr^{3+}(aq)$ in the cell.

34. A galvanic cell is constructed in which the overall reaction is

$$Cr_2O_7^{2-}(aq) + 14\ H_2O^+(aq) + 6\ I^-(aq) \longrightarrow 2\ Cr^{3+}(aq) + 3\ I_2(s) + 21\ H_2O(\ell)$$

(a) Calculate E^o for this cell.
(b) At pH 0, with $[Cr_2O_7^{2-}] = 1.5$ M and $[I^-] = 0.40$ M, the cell potential is found to equal 0.87 V. Calculate the concentration of $Cr^{3+}(aq)$ in the cell.

35. By using the half-cell potentials in Appendix E, calculate the equilibrium constant at 25°C for the reaction in problem 33. Dichromate ion $(Cr_2O_7^{2-})$ is orange, and Cr^{3+} is light green in aqueous solution. If 2.00 L of 1.00 M $HClO_2$ solution is added to 2.00 L of 0.50 M $Cr(NO_3)_3$ solution, what color will the resulting solution have?

36. By using the half-cell potentials in Appendix E, calculate the equilibrium constant at 25°C for the reaction

$$6\ Hg^{2+}(aq) + 2\ Au(s) \rightleftharpoons 3\ Hg_2^{2+}(aq) + 2\ Au^{3+}(aq)$$

If 1.00 L of a 1.00 M $Au(NO_3)_3$ solution is added to 1.00 L of a 1.00 M $Hg_2(NO_3)_2$ solution, calculate the concentrations of Hg^{2+}, Hg_2^{2+}, and Au^{3+} at equilibrium.

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37. The following standard reduction potentials have been determined for the aqueous chemistry of indium:

$$In^{3+}(aq) + 2\,e^- \longrightarrow In^+(aq) \qquad E^o = -0.40\text{ V}$$

$$In^+(aq) + e^- \longrightarrow In(s) \qquad E^o = -0.21\text{ V}$$

Calculate the equilibrium constant (K) for the disproportionation of $In^+(aq)$ at 25°C.

$$3\,In^+(aq) \rightleftharpoons 2\,In(s) + In^{3+}(aq)$$

38. Use data from Appendix E to compute the equilibrium constant for the reaction

$$Hg^{2+}(aq) + Hg(\ell) \rightleftharpoons Hg_2^{2+}(aq)$$

39. A galvanic cell consists of a $Pt\,|\,H_3O^+(1.00\text{ M})\,|\,H_2(g)$ cathode connected to a $Pt\,|\,H_3O^+(aq)\,|\,H_2(g)$ anode in which the concentration of H_3O^+ is unknown but is kept constant by the action of a buffer consisting of a weak acid, HA(0.10 M), mixed with its conjugate base, A^-(0.10 M). The measured cell potential is $E_{cell} = 0.150$ V at 25°C, with a hydrogen pressure of 1.00 atm at both electrodes. Calculate the pH in the buffer solution, and from it determine the K_a of the weak acid.

40. In a galvanic cell, the cathode consists of a $Ag^+(1.00\text{ M})\,|\,Ag$ half-cell. The anode is a platinum wire, with hydrogen bubbling over it at 1.00-atm pressure, which is immersed in a buffer solution containing benzoic acid and sodium benzoate. The concentration of benzoic acid (C_6H_5COOH) is 0.10 M, and that of benzoate ion ($C_6H_5COO^-$) is 0.050 M. The overall cell reaction is then

$$Ag^+(aq) + \frac{1}{2}H_2(g) + H_2O(\ell) \longrightarrow Ag(s) + H_3O^+(aq)$$

and the measured cell potential is 1.030 V. Calculate the pH in the buffer solution and determine the K_a of benzoic acid.

41. A galvanic cell is constructed in which the overall reaction is

$$Br_2(\ell) + H_2O(g) + 2\,H_2O(\ell) \longrightarrow 2\,Br^-(aq) + 2\,H_3O^+(aq)$$

(a) Calculate E^o for this cell.
(b) Silver ions are added until AgBr precipitates at the cathode and $[Ag^+]$ reaches 0.060 M. The cell potential is then measured to be 1.710 V at pH = 0 and = 1.0 atm. Calculate $[Br^-]$ under these conditions.
(c) Calculate the solubility product constant K_{sp} for AgBr.

42. A galvanic cell is constructed in which the overall reaction is

$$Pb(s) + 2\,H_3O^+(aq) \longrightarrow Pb^{2+}(aq) + H_2(g) + 2\,H_2O(\ell)$$

(a) Calculate E^{os} for this cell.
(b) Chloride ions are added until $PbCl_2$ precipitates at the anode and $[Cl^-]$ reaches 0.15 M. The cell potential is then measured to be 0.22 V at pH = 0 and P_{H_2} = 1.0 atm. Calculate $[Pb^{2+}]$ under these conditions.
(c) Calculate the solubility product constant K_{sp} of $PbCl_2$.

Molecular Electrochemistry

43. Sketch and interpret the cyclic voltammogram you expect for the $[Ru(NH_3)_6]^{3+}/[Ru(NH_3)_6]^{2+}$ redox couple measured with a Ag/AgCl reference electrode. The standard reduction potential for this system is −0.215 V measured with respect to SCE.

44. Ferrocene is an organometallic complex that contains iron sandwiched between two five-membered cyclopentadiene rings (C_5H_5). Ferrocene methanol, a derivative, is another convenient reference redox couple used in electrochemistry. Sketch and interpret the cyclic voltammogram for the ferrocene methanol couple (Fc^+/Fc) measured with respect to SHE; the standard redox potential is 0.64 V.

45. What is the wavelength and color of light emitted in an ECL experiment in which the redox potential associated with the HOMO is 1.5 V and that associated with the LUMO is −0.9 V? The experiment is conducted in a nonaqueous solution.

46. Suppose that you were asked to synthesize a molecule for use in ECL that emits light at 520 nm and has a potential associated with the HOMO located at 0.5 V, for compatibility with other requirements in a particular analytical application. What is the redox potential associated with the LUMO?

47. Would CdS be a suitable semiconductor for direct photoelectrochemical water splitting? Why or why not? The conduction band lies at about −1.25 V vs NHE and the valence band lies at about 0.12 V vs NHE.

48. GaP has a bandgap of 2.3 eV with a conduction band redox potential located at −1.3 V vs SHE. Would this material be suitable for direct photoelectrochemical water splitting? Why or why not?

49. Strontium titanate ($SrTiO_3$) is another wide bandgap semiconductor that has been investigated for use in dye-sensitized solar cells. Depending on morphology and pH the conduction band of $SrTiO_3$ lies around −0.2 V vs. SHE. What energies for the HOMO and LUMO of a sensitizing dye would you specify for electron injection into the conduction band of strontium titanate and for water oxidation by the radical cation formed by photoexcitation?

50. A Grätzel cell is to be designed using strontium titanate as the semiconductor and the iodide/triode mediator discussed in the *Connection to Energy:* Solar Energy Conversion. Where should the energy levels of the sensitizer lie for maximum solar-to-electrical conversion efficiency?

Batteries and Fuel Cells

51. Calculate the potential E^o of a lead–acid cell if all reactants and products are in their standard states. What will be the voltage if six such cells are connected in a series?

52. Calculate the standard potential of the zinc–mercuric oxide cell shown in Figure 17.22. (*Hint:* The easiest way to proceed is to calculate ΔG° for the corresponding overall reaction, and then find E^o from it.) Take Δ_f^o $[Zn(OH)_2(s)] = -553.5\text{ kJ mol}^{-1}$.

53. (a) What quantity of charge (in coulombs) is a fully charged 12-V lead–acid storage battery theoretically capable of furnishing if the spongy lead available for reaction at the anodes weighs 10 kg and there is excess PbO_2?
(b) What is the theoretical maximum amount of work (in joules) that can be obtained from this battery?

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54. (a) What quantity of charge (in coulombs) is a fully charged 1.34-V zinc–mercuric oxide watch battery theoretically capable of furnishing if the mass of HgO in the battery is 0.50 g?
(b) What is the theoretical maximum amount of work (in joules) that can be obtained from this battery?

55. The concentration of the electrolyte, sulfuric acid, in a lead–acid storage battery diminishes as the battery is discharged. Is a discharged battery recharged by replacing the dilute H_2SO_4 with fresh, concentrated H_2SO_4? Explain.

56. One cold winter morning the temperature is well below 0°F. In trying to start your car, you run the battery down completely. Several hours later, you return to replace your fouled spark plugs and find that the liquid in the battery has now frozen even though the air temperature is actually a bit higher than it was in the morning. Explain how this can happen.

57. Consider the fuel cell that accomplishes the overall reaction

$$H_2(g) + \frac{1}{2}O_2(g) \longrightarrow H_2O(\ell)$$

If the fuel cell operates with 60% efficiency, calculate the amount of electrical work generated per gram of water produced. The gas pressures are constant at 1 atm, and the temperature is 25°C.

58. Consider the fuel cell that accomplishes the overall reaction

$$CO(g) + \frac{1}{2}O_2(g) \longrightarrow CO_2(g)$$

Calculate the maximum electrical work that could be obtained from the conversion of 1.00 mol of $CO(g)$ to $CO_2(g)$ in such a fuel cell operated with 100% efficiency at 25°C and with the pressure of each gas equal to 1 atm.

Corrosion and Corrosion Prevention

59. Two half-reactions proposed for the corrosion of iron in the absence of oxygen are

$$Fe(s) \longrightarrow Fe^{2+}(aq) + 2\,e^-$$

$$2\,H_2O(\ell) + 2\,e^- \longrightarrow 2\,OH^-(aq) + H_2(g)$$

Calculate the standard cell potential generated by a galvanic cell running this pair of half-reactions. Is the overall reaction spontaneous under standard conditions? As the pH falls from 14, will the reaction become spontaneous?

60. In the presence of oxygen, the cathode half-reaction written in the preceding problem is replaced by

$$\frac{1}{2}O_2(g) + 2\,H_3O^+(aq) + 2\,e^- \longrightarrow 3\,H_2O(\ell)$$

but the anode half-reaction is unchanged. Calculate the standard cell potential for *this* pair of reactions operating as a galvanic cell. Is the overall reaction spontaneous under standard conditions? As the water becomes more acidic, does the driving force for the rusting of iron increase or decrease?

61. Could sodium be used as a sacrificial anode to protect the iron hull of a ship?

62. If it is shown that titanium can be used as a sacrificial anode to protect iron, what conclusion can be drawn about the standard reduction potential of its half-reaction?

$$Ti^{3+}(aq) + 3\,e^- \longrightarrow Ti(s)$$

Electrometallurgy

63. In the Downs process, molten sodium chloride is electrolyzed to produce sodium. A valuable byproduct is chlorine. Write equations representing the processes taking place at the anode and at the cathode in the Downs process.

64. The first element to be prepared by electrolysis was potassium. In 1807, Humphry Davy, then 29 years old, passed an electric current through molten potassium hydroxide (KOH), obtaining liquid potassium at one electrode and water and oxygen at the other. Write equations to represent the processes taking place at the anode and at the cathode.

65. A current of 55,000 A is passed through a series of 100 Hall–Héroult cells for 24 hours. Calculate the maximum theoretical mass of aluminum that can be recovered.

66. A current of 75,000 A is passed through an electrolysis cell containing molten $MgCl_2$ for 7.0 days. Calculate the maximum theoretical mass of magnesium that can be recovered.

67. An important use for magnesium is to make titanium. In the Kroll process, magnesium reduces titanium(IV) chloride to elemental titanium in a sealed vessel at 800°C. Write a balanced chemical equation for this reaction. What mass of magnesium is needed, in theory, to produce 100 kg of titanium from titanium(IV) chloride?

68. Calcium is used to reduce vanadium(V) oxide to elemental vanadium in a sealed steel vessel. Vanadium is used in vanadium steel alloys for jet engines, high-quality knives, and tools. Write a balanced chemical equation for this process. What mass of calcium is needed, in theory, to produce 20.0 kg of vanadium from vanadium(V) oxide?

69. Galvanized steel consists of steel with a thin coating of zinc to reduce corrosion. The zinc can be deposited electrolytically, by making the steel object the cathode and a block of zinc the anode in an electrochemical cell containing a dissolved zinc salt. Suppose a steel garbage can is to be galvanized and requires that a total mass of 7.32 g of zinc be coated to the required thickness. How long should a current of 8.50 A be passed through the cell to achieve this?

70. In the electroplating of a silver spoon, the spoon acts as the cathode and a piece of pure silver as the anode. Both dip into a solution of silver cyanide (AgCN). Suppose that a current of 1.5 A is passed through such a cell for 22 minutes and that the spoon has a surface area of 16 cm^2. Calculate the average thickness of the silver layer deposited on the spoon, taking the density of silver to be 10.5 g cm^{-3}.

A Deeper Look . . . Electrolysis of Water and Aqueous Solutions

71. An electrolytic cell consists of a pair of inert metallic electrodes in a solution buffered to pH = 5.0 and containing nickel sulfate ($NiSO_4$) at a concentration of 1.00 M. A current of 2.00 A is passed through the cell for 10.0 hours.

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(a) What product is formed at the cathode?
(b) What is the mass of this product?
(c) If the pH is changed to pH = 1.0, what product will form at the cathode?

72. A 0.100 M neutral aqueous $CaCl_2$ solution is electrolyzed using platinum electrodes. A current of 1.50 A passes through the solution for 50.0 hours.
(a) Write the half-reactions occurring at the anode and at the cathode.
(b) What is the decomposition potential?
(c) Calculate the mass, in grams, of the product formed at the cathode.

ADDITIONAL PROBLEMS

73. The drain cleaner Drano consists of aluminum turnings mixed with sodium hydroxide. When it is added to water, the sodium hydroxide dissolves and releases heat. The aluminum reacts with water to generate bubbles of hydrogen and aqueous ions. Write a balanced net ionic equation for this reaction.

74. Sulfur-containing compounds in the air tarnish silver, giving black Ag_2S. A practical method of cleaning tarnished silverware is to place the tarnished item in electrical contact with a piece of zinc and dip both into water containing a small amount of salt. Write balanced half-equations to represent what takes place.

75. A current passed through inert electrodes immersed in an aqueous solution of sodium chloride produces chlorate ion, $ClO_3^-(aq)$, at the anode and gaseous hydrogen at the cathode. Given this fact, write a balanced equation for the chemical reaction if gaseous hydrogen and aqueous sodium chlorate are mixed and allowed to react spontaneously until they reach equilibrium.

76. A galvanic cell is constructed by linking a $Co^{2+}|Co(s)$ half-cell to an $Ag^+|Ag(s)$ half-cell through a salt bridge and then connecting the cobalt and silver electrodes through an external circuit. When the circuit is closed, the cell potential is measured to be 1.08 V, and silver is seen to plate out while cobalt dissolves.
(a) Write the half-reactions that occur at the anode and at the cathode and the balanced overall cell reaction.
(b) The cobalt electrode is weighed after 150 minutes of operation and is found to have decreased in mass by 0.36 g. By what amount has the silver electrode increased in mass?
(c) What is the average current drawn from the cell during this period?

77. The galvanic cell $Zn(s)|Zn^{2+}(aq)||Ni^{2+}(aq)|Ni(s)$ is constructed using a completely immersed zinc electrode that weighs 32.68 g and a nickel electrode immersed in 575 mL of 1.00 M $Ni^{2+}(aq)$ solution. A steady current of 0.0715 A is drawn from the cell as the electrons move from the zinc electrode to the nickel electrode.
(a) Which reactant is the limiting reactant in this cell?
(b) How long does it take for the cell to be completely discharged?
(c) How much mass has the nickel electrode gained when the cell is completely discharged?
(d) What is the concentration of the $Ni^{2+}(aq)$ when the cell is completely discharged?

78. A newly discovered bacterium can reduce selenate ion, $SeO_4^{2-}(aq)$, to elemental selenium, $Se(s)$, in reservoirs. This is significant because the soluble selenate ion is potentially toxic, but elemental selenium is insoluble and harmless. Assume that water is oxidized to oxygen as the selenate ion is reduced. Compute the mass of oxygen produced if all the selenate in a 1012-L reservoir contaminated with 100 mg L^{-1} of selenate ion is reduced to selenium.

79. Thomas Edison invented an electric meter that was nothing more than a simple coulometer, a device to measure the amount of electricity passing through a circuit. In this meter, a small, fixed fraction of the total current supplied to a household was passed through an electrolytic cell, plating out zinc at the cathode. Each month the cathode could then be removed and weighed to determine the amount of electricity used. If 0.25% of a household's electricity passed through such a coulometer and the cathode increased in mass by 1.83 g in a month, how many coulombs of electricity were used during that month?

80. The chief chemist of the Brite-Metal Electroplating Co. is required to certify that the rinse solutions that are discharged from the company's tin-plating process into the municipal sewer system contain no more than 10 ppm (parts per million) by mass of Sn^{2+}. The chemist devises the following analytical procedure to determine the concentration. At regular intervals, a 100-mL (100-g) sample is withdrawn from the waste stream and acidified to pH = 1.0. A starch solution and 10 mL of 0.10 M potassium iodide are added, and a 25.0-mA current is passed through the solution between platinum electrodes. Iodine appears as a product of electrolysis at the anode when the oxidation of Sn^{2+} to Sn^{4+} is practically complete and signals its presence with the deep blue color of a complex formed with starch. What is the maximum duration of electrolysis to the appearance of the blue color that ensures that the concentration of Sn^{2+} does not exceed 10 ppm?

81. Estimate the cost of the electrical energy needed to produce 1.5×10^{10} kg (a year's supply for the world) of aluminum from $Al_2O_3(s)$ if electrical energy costs 10 cents per kilowatt-hour (1 kWh = 3.6 MJ = 3.6×10^6 J) and if the cell potential is = V.

82. Titanium can be produced by electrolytic reduction from an anhydrous molten salt electrolyte that contains titanium(IV) chloride and a spectator salt that furnishes ions to make the electrolyte conduct electricity. The standard enthalpy of formation of $TiCl_4(l)$ is -750 kJ mol^{-1}, and the standard entropies of $TiCl_4(l)$, $Ti(s)$, and $Cl_2(g)$ are 253, 30, and 223 J K^{-1} mol^{-1}, respectively. What minimum applied voltage will be necessary at 100°C?

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83. A half-cell has a graphite electrode immersed in an acidic solution (pH 0) of Mn^{2+} (concentration 1.00 M) in contact with solid MnO_2. A second half-cell has an acidic solution (pH 0) of H_2O_2 (concentration 1.00 M) in contact with a platinum electrode past which gaseous oxygen at a pressure of 1.00 atm is bubbled. The two half-cells are connected to form a galvanic cell.
   (a) Referring to Appendix E, write balanced chemical equations for the half-reactions at the anode and the cathode and for the overall cell reaction.
   (b) Calculate the cell potential.

84. By considering these half-reactions and their standard reduction potentials,

$$Pt^{2+} + 2\,e^- \longrightarrow Pt \qquad E^o = 1.2\text{ V}$$

$$NO_3^- + 4\,H_3O + 3\,e^- \longrightarrow NO + 6\,H_2O \qquad E^o = 0.96\text{ V}$$

$$PtCl_4^{2-} + 2\,e^- \longrightarrow Pt + 4\,Cl^- \qquad E^o = 0.73\text{ V}$$

account for the fact that platinum will dissolve in a mixture of hydrochloric acid and nitric acid (*aqua regia*) but will not dissolve in either acid alone.

85. (a) One method to reduce the concentration of unwanted $Fe^{3+}(aq)$ in a solution of $Fe^{2+}(aq)$ is to drop a piece of metallic iron into the storage container. Write the reaction that removes the Fe^{3+}, and compute its standard cell potential.
   (b) Referring to problem 23, suggest a way to remove unwanted $Mn^{3+}(aq)$ from solutions of $Mn^{2+}(aq)$.

86. (a) Based only on the standard reduction potentials for the $Cu^{2+}|Cu^+$ and $I_2(s)|I^-$ the half-reactions, would you expect $Cu^{2+}(aq)$ to be reduced to $Cu^+(aq)$ by $I^-(aq)$?
   (b) The formation of solid CuI plays a role in the interaction between $Cu^{2+}(aq)$ and $I^-(aq)$.

$$Cu^{2+}(aq) + I^-(aq) + e^- \rightleftharpoons CuI(s) \qquad E^o = 0.86\text{ V}$$

Taking into account this added information, do you expect Cu^{2+} to be reduced by iodide ion?

87. In some old European churches, the stained-glass windows have so darkened from corrosion and age that hardly any light comes through. Microprobe analysis showed that tiny cracks and defects on the glass surface were enriched in insoluble Mn(III) and Mn(IV) compounds. From Appendix E, suggest a reducing agent and conditions that might successfully convert these compounds to soluble Mn(II) without simultaneously reducing Fe(III) (which gives the glass its colors) to Fe(II). Take MnO_2 as representative of the insoluble Mn(III) and Mn(IV) compounds.

88. (a) Calculate the half-cell potential for the reaction

$$O_2(g) + 4\,H_3O^+(aq) + 4\,e^- \longrightarrow 6\,H_2O(\ell)$$

at pH 7 with the oxygen pressure at 1 atm.
   (b) Explain why aeration of solutions of I^- leads to their decomposition. Write a balanced equation for the redox reaction that occurs.
   (c) Will the same problem arise with solutions containing Br^- or Cl^-? Explain.
   (d) Will decomposition be favored or opposed by increasing acidity?

89. An engineer needs to prepare a galvanic cell that uses the reaction

$$2\,Ag^+(aq) + Zn(s) \longrightarrow Zn^{2+}(aq) + 2\,Ag(s)$$

and generates an initial voltage of 1.50 V. She has 0.010 M $AgNO_3(aq)$ and 0.100 M $Zn(NO_3)_2(aq)$ solutions, as well as electrodes of metallic copper and silver, wires, containers, water, and a KNO_3 salt bridge. Sketch the cell. Clearly indicate the concentrations of all solutions.

90. Consider a galvanic cell for which the anode reaction is

$$Pb(s) \longrightarrow Pb^{2+}(1.0 \times 10^{-2}\text{ M}) + 2\,e^-$$

and the cathode reaction is

$$VO^{2+}(0.10\text{ M}) + 2\,H_3O^+(0.10\text{ M}) + e^- \longrightarrow V^{3+}(1.0 \times 10^{-5}\text{ M}) + 3\,H_2O(\ell)$$

The measured cell potential is 0.640 V.
   (a) Calculate E^o for the $VO^{2+}|V^{3+}$ half-reaction, using $E^o(Pb^{2+}|Pb)$ from Appendix E.
   (b) Calculate the equilibrium constant (K) at 25°C for the reaction

$$Pb(s) + 2\,VO^{2+}(aq) + 4\,H_3O^+(aq) \rightleftharpoons Pb^{2+}(aq) + 2\,V^{3+}(aq) + 6\,H_2O(\ell)$$

91. Suppose we construct a pressure cell in which the gas pressures differ in the two half-cells. Suppose such a cell consists of a $Cl_2(0.010\text{ atm})|Cl^-(1\text{ M})$ half-cell connected to a $Cl_2(0.50\text{ atm})|Cl^-(1\text{ M})$ half-cell. Determine which half-cell will be the anode, write the overall equation for the reaction, and calculate the cell potential.

92. A student decides to measure the solubility of lead sulfate in water and sets up the electrochemical cell

$$Pb|PbSO_4|SO_4^{2-}(aq, 0.0500\text{ M})||Cl^-(aq, 1.00\text{ M})|AgCl|Ag$$

At 25°C the student finds the cell potential to be 0.546 V, and from Appendix E the student finds

$$AgCl(s) + e \rightleftharpoons Ag(s) + Cl^-(aq) \qquad E^o = 0.222\text{ V}$$

What does he find for the K_{sp} of $PbSO_4$?

93. A wire is fastened across the terminals of the Leclanché cell in Figure 17.21. Indicate the direction of electron flow in the wire.

94. Overcharging a lead–acid storage battery can generate hydrogen. Write a balanced equation to represent the reaction taking place.

95. An ambitious chemist discovers an alloy electrode that is capable of catalytically converting ethanol reversibly to carbon dioxide at 25°C according to the half-reaction

$$C_2H_5OH(\ell) + 15\,H_2O(\ell) \longrightarrow 2\,CO_2(g) + 12\,H_3O^+(aq) + 12\,e^-$$

Believing that this discovery is financially important, the chemist patents its composition and designs a fuel cell that may be represented as

$$\text{Alloy}|C_2H_5OH(\ell)|CO_2(g) + H_3O^+(1\text{ M})||H_3O^+(1\text{ M})|O_2|Ni$$

   (a) Write the half-reaction occurring at the cathode.

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(b) Using data from Appendix D, calculate $E°$ for the cell at 25°C.
(c) What is the $E°$ value for the ethanol half-cell?

96. Iron or steel is often covered by a thin layer of a second metal to prevent rusting: Tin cans consist of steel covered with tin, and galvanized iron is made by coating iron with a layer of zinc. If the protective layer is broken, however, iron will rust more readily in a tin can than in galvanized iron. Explain this observation by comparing the half-cell potentials of iron, tin, and zinc.

97. An electrolysis cell contains a solution of 0.10 M $NiSO_4$. The anode and cathode are both strips of Pt foil. Another electrolysis cell contains the same solution, but the electrodes are strips of Ni foil. In each case a current of 0.10 A flows through the cell for 10 hours.
(a) Write a balanced equation for the chemical reaction that occurs at the anode in each cell.
(b) Calculate the mass, in grams, of the product formed at the anode in each cell. (The product may be a gas, a solid, or an ionic species in solution.)

98. A potential difference of 2.0 V is impressed across a pair of inert electrodes (e.g., platinum) that are immersed in a 0.050 M aqueous KBr solution. What are the products that form at the anode and the cathode?

99. An aqueous solution is simultaneously 0.10 M in $SnCl_2$ and in $CoCl_2$.
(a) If the solution is electrolyzed, which metal will appear first?
(b) At what decomposition potential will that metal first appear?
(c) As the electrolysis proceeds, the concentration of the metal being reduced will drop and the potential will change. How complete a separation of the metals using electrolysis is theoretically possible? In other words, at the point where the second metal begins to form, what fraction of the first metal is left in solution?

100. A 55.5-kg slab of crude copper from a smelter has a copper content of 98.3%. Estimate the time required to purify it electrochemically if it is used as the anode in a cell that has acidic copper(II) sulfate as its electrolyte and a current of 2.00×10^3 A is passed through the cell.

101. Sheet iron can be galvanized by passing a direct current through a cell containing a solution of zinc sulfate between a graphite anode and the iron sheet. Zinc plates out on the iron. The process can be made continuous if the iron sheet is a coil that unwinds as it passes through the electrolysis cell and coils up again after it emerges from a rinse bath. Calculate the cost of the electricity required to deposit a 0.250-mm-thick layer of zinc on both sides of an iron sheet that is 1.00 m wide and 100 m long, if a current of 25 A at a voltage of 3.5 V is used and the energy efficiency of the process is 90%. The cost of electricity is 10 cents per kilowatt–hour (1 kWh = 3.6 MJ). Consult Appendix F for data on zinc.

CUMULATIVE PROBLEMS

102. A 1.0 M solution of NaOH is electrolyzed, generating $O_2(g)$ at the anode. A current of 0.15 A is passed through the cell for 75 minutes. Calculate the volume of (wet) oxygen generated in this period if the temperature is held at 25°C and the total pressure is 0.985 atm. (*Hint:* Use the vapor pressure of water at this temperature from Table 10.3.)

103. Use standard entropies from Appendix D to predict whether the standard potential of the $Cu|Cu^{2+}\|Ag^+|Ag$ cell (diagrammed in Figure 17.2) will increase or decrease if the temperature is raised above 25°C.

104. About 50,000 kJ of electrical energy is required to produce 1.0 kg of Al from its $Al(OH)^3$ ore. The major energy cost in recycling aluminum cans is the melting of the aluminum. The enthalpy of fusion of Al(*s*) is 10.7 kJ mol^{-1}. Compare the energy cost for making new aluminum with that for recycling.

105. (a) Use the following half-reactions and their reduction potentials to calculate the K_{sp} of AgBr:

$$Ag^+ + e^- \longrightarrow Ag(s) \qquad E^o = 0.7996 \text{ V}$$

$$AgBr(s) + e^- \longrightarrow Ag(s) + Br^- \qquad E^o = 0.0713 \text{ V}$$

(b) Estimate the solubility of AgBr in 0.10 M NaBr(*aq*).

106. Amounts of iodine dissolved in aqueous solution, $I_2(aq)$, can be determined by titration with thiosulfate ion ($S_2O_3^{2-}$). The thiosulfate ion is oxidized to $S_4O_6^{2-}$ while the iodine is reduced to iodide ion. Starch is used as an indicator because it has a strong blue color in the presence of dissolved iodine.
(a) Write a balanced equation for this reaction.
(b) If 56.40 mL of 0.100 M $S_2O_3^{2-}$ solution is used to reach the endpoint of a titration of an unknown amount of iodine, calculate the number of moles of iodine originally present.
(c) Combine the appropriate half-cell potentials from Appendix E with thermodynamic data from Appendix D for the equilibrium

$$I_2(s) \rightleftharpoons I_2(aq)$$

to calculate the equilibrium constant at 25°C for the reaction in part (a).

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